

c) The queuing model II notation as per Kendall and Lee notation is given as:

$$M/M/1 : N/FCFS.$$

Explanation

M: This indicates that arrivals follows a poisson process/distribution.

M: This indicates that Service time follows exponential distribution.

1: This indicates there is only one server handling all arriving customers.

N: Indicates that the system capacity is finite. Once the system has N customers, any new arrivals are blocked or lost.

FCFS (First-Come-First-Served): This define the queuing discipline, where customers are served in the order they arrive.

Data given,

b)

Number of Servers $N=2$.

Arrival rate $\lambda = 15$ calls/hour.

Average Service time = 5 minutes $\rightarrow$ Service rate.

$$\mu = \frac{1}{5} \times 60 = 12 \text{ Calls / hour per Server.}$$

(2) Probability that a customer should have to wait for his long distant calls on the peak hour of the day.
for this case

$$P_{\text{wait}} = 1 - P_0 - P_1$$

b) For P_0

$$P_0 = \frac{1}{\left[\sum_{n=0}^{N-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{N!} \left(\frac{\lambda}{\mu} \right)^N \cdot \frac{N\mu}{N\mu - \lambda} \right]}$$

Where

P_0 = Probability that no customer are in the system [idle system].

λ = Arrival rate.

μ = Service rate per server.

N = Number of server.

n = Number of customers in the system.

When $n < N$ All customers are being served.

$n \geq N$ A queue forms because all servers are busy.

$$P_0 = \frac{1}{\left[\frac{1}{0!} \left(\frac{15}{12} \right)^0 + \frac{1}{1!} \left(\frac{15}{12} \right)^1 + \left[\frac{1}{2!} \left(\frac{15}{12} \right)^2 \times \frac{2 \times 12}{2 \times 12 - 15} \right] \right]}$$

$$P_0 = \frac{1}{1 + 1.25 + 0.78125 \times 2.66667}$$

$$P_0 = 0.2308.$$

For P_1

from

$$P_n = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n \cdot P_0$$

$$P_1 = \frac{1}{1!} \left(\frac{15}{12} \right)^1 \times 0.2308.$$

$$P_1 = 1.25 \times 0.2308$$

(i) Now

$$P_1 = 0.2885,$$

$$P_{wait} = 1 - 0.2308 - 0.2885,$$

$$P_{wait} = 0.4807.$$

∴ For The probability that a customer will have to wait is

$$\underline{\underline{0.4807}}$$

ii) Expected number of customers in the queue (L_q).

For this case

$$L_q = \frac{\lambda \mu \left(\frac{\lambda}{\mu}\right)^N}{(N-1)! \cdot (N\mu - \lambda)^2} \times P_0.$$

Where

N = Number of Servers,

λ = Arrival rate,

μ = Service rate per server,

P_0 = Probability that the System is empty.

$$L_q = \frac{15 \times 12 \left(\frac{15}{12}\right)^2}{(2-1)! \times (2 \times 12 - 15)^2} \times 0.2308$$

$$L_q = \frac{281.25}{81} \times 0.2308$$

$$L_q = 0.797$$

Customers in the queue is 0.797 customers.

b iii) Average waiting time for the customers,
from

$$W_q = \frac{L_q}{\lambda}$$

$$W_q = \frac{0.797}{15}$$

$$W_q = 0.053 \text{ hours, } = 3.18 \text{ minutes,}$$

∴ Average waiting time is ≈ 0.053 hours,